Presentation Script

Group 3 · Can AI Foundation Models Beat Traditional Statistical Models? · Oct 21, 2026 · @Shihan

Run of show

The full script runs about 11 minutes, including the 90-second demo at the end. Sentences marked *(optional)* add up to about 40 seconds: skip them to finish in 10 minutes without losing a finding. The same script is in each slide's speaker notes in the deck. Swap speaker names as the group prefers.

Slide	Title	Speaker	Time	Ends at	The one message
1	Can AI foundation models beat traditional statistical models?	Speaker 1	0:45	0:45	The question: accuracy and efficiency
2	5,748 daily series, one 28-day test window	Speaker 1	1:20	2:05	Hard data; the same unseen 28 days for every model
3	Five models plus a simple average	Speaker 2	1:05	3:10	Statistics, machine learning, zero-shot AI, a benchmark
4	Traditional wins accuracy; AI wins speed	Speaker 2	1:05	4:15	Traditional methods lead; their average is best
5	Traditional tracks the peaks; the AI runs low	Speaker 2	1:10	5:25	Gaps are at store level; Chronos is by far the fastest
6	Claude: a language model that reads sales as text	Speaker 3	1:15	6:40	Claude matched the best on totals; how totals are built matters
7	Using an LLM to forecast: risks and safeguards	Speaker 3	0:55	7:35	The blind test can't fully rule out contamination
8	Our answer: not yet on accuracy, yes on speed	Speaker 4	2:00	9:35	Answer, limitations, newsvendor insight, recommendation
9	Demo: Shelf Signal dashboard	Speaker 4	1:45	11:20	90-second demo, then thanks and questions

Delivery tips: speak at about 130 words a minute; point at one thing per chart; use the transition line to hand over. Rehearse once with a timer. If you need to finish in 10 minutes, skip the sentences marked optional; never cut the demo below 60 seconds.

Speaker 1 — question and data (2:05)

Slide 1 · Can AI foundation models beat traditional statistical models? (0:45)

> Good afternoon, we're Group 3. A retail forecast has three jobs: be accurate, be cheap and fast enough to run every day for thousands of products, and support good stocking decisions. LightGBM, a machine-learning method, dominates this kind of retail data: it dominated Walmart's M5 forecasting competition and became the industry default. Since then, AI foundation models, pre-trained on huge datasets, can forecast a product they have never seen, with no training at all. So we ask: can these AI models beat traditional statistical and machine-learning methods, on accuracy and on efficiency?

Transition: "Here is our data and how we tested."

Slide 2 · 5,748 daily series, one 28-day test window (1:20)

> We used Walmart's M5 data: all 1,437 food items in four California stores, so 5,748 daily series, each one item in one store, from 2011 to 2016.
>
> Every model trains on data up to May 22, 2016, and forecasts the next 28 days, the official M5 test window, which no model sees. The one model we tuned, LightGBM, was tuned only on the last 28 training days.
>
> Three features make this data hard. Sales follow a weekly rhythm: weekends sell about 20% above an average day. Food-stamp payment days add about 10%. And demand is sparse: the typical item sells nothing on half of all days.
>
> We score accuracy with WRMSSE, the competition's own metric: it scales each product's error by how hard it is to forecast and weights it by dollar sales. Lower is better, and below one beats a naive forecast. One caveat up front: this is a single 28-day window.

Point at: the amber "Test: 28 days" bar.

Transition: "Speaker 2 will introduce the models."

Speaker 2 — models and results (3:05)

Slide 3 · Five models plus a simple average (1:05)

> We compared five models. AutoETS and AutoARIMA are classic time series models: they fit one model to each product's own history, with a weekly cycle.
>
> LightGBM is the dominant approach for this kind of retail data; it dominated the M5 competition. It is one machine-learning model trained across all products at once, which then predicts each product. It also sees prices, holidays and food-stamp days.
>
> Chronos is Amazon's pre-trained AI model for time series, trained on about 100 billion data points from energy, traffic, weather and more. It was not trained on M5 or on any retail data: the Chronos paper lists M5 as a held-out test set. Seasonal naive, "same weekday last week", is the benchmark.
>
> We also test the simplest possible combination: an unweighted average of AutoETS and LightGBM. And as an experiment, a second AI joins on a subset of series: Claude, a general-purpose language model, which we'll come back to later. Note that the inputs differ: only LightGBM and Claude see prices and the calendar.

Transition: "So which model is best, and at what? Here's the scorecard."

Slide 4 · Traditional wins accuracy; AI wins speed (1:00)

> Here is the scorecard: traditional models against AI, on four questions. Who is most accurate overall? A traditional model: the simple average of AutoETS and LightGBM, with LightGBM alone second and AutoETS close behind. Who forecasts store totals best? Again the traditional models. For single products? It's close, and there the AI is level with LightGBM. And who is fastest? The AI, Chronos, by far: seconds instead of minutes, with no training at all.
>
> So from a traditional point of view, the best choice is the AutoETS-plus-LightGBM average, and the AI's real strength is speed. The green cells show the winner of each column.

Point at: the green cells, column by column.

Transition: "Let's see what those forecasts actually look like."

Slide 5 · Traditional tracks the peaks; the AI runs low (1:00)

> Here's what that looks like. The black line is actual daily sales across all four stores during the test month. The traditional average, in teal, rises and falls with every weekend peak. AutoARIMA, in green, is almost flat, and Chronos, in orange, has the right shape but sits too low.
>
> Why? AutoARIMA and Chronos look at one product at a time. For a product that sells a unit or two a day, the best guess is a flat line, and thousands of flat lines added up miss the weekend. LightGBM learns from all products together, so it sees the weekend rush, and averaging it with AutoETS smooths out the rest. Zoom into a single product, though, and the lines come together: one product's daily sales are mostly noise.

Transition: "So the specialist AI is fast but less accurate. What about a generalist AI that reads and reasons? Speaker 3."

Speaker 3 — the LLM experiment (2:10)

Slide 6 · Claude: a language model that reads sales as text (1:15)

> Now a generalist AI. We used Claude through its API. For each product, our code sends the past year of daily sales as plain text, one row per day with the date, holidays, food-stamp days and price, plus an instruction: forecast the next 28 days and answer in a fixed format. Claude reads the numbers, reasons about patterns such as weekends and holidays, and sends back 28 daily forecasts with a short explanation. No statistical model is fitted; it works from what it learned in training.
>
> Each request costs money and takes a few seconds, so to keep the cost down we used a subset instead of all 5,748 series: 100 randomly chosen products and the 12 store-by-department totals. Every other model was scored on exactly the same series, and to keep it fair, the other models were also fitted directly on those totals.
>
> How did it perform? On the store totals, very well: it matched our best traditional forecast, the AutoETS-plus-LightGBM average, and beat every single model, including Chronos. On individual products it was in the pack, behind AutoETS and AutoARIMA. So the generalist AI was more accurate than the specialist one, but far slower and more costly to run at scale.

Point at: the Chronos vs Claude table.

Transition: "Using a language model also brings specific risks."

Slide 7 · Using an LLM to forecast: risks and safeguards (1:00)

> The obvious worry: M5 is public data, so Claude may have seen these exact sales during training, and be remembering rather than forecasting. So we ran a blind test. Every forecast was made twice: once with the real labels, meaning Walmart, the store and product IDs, and the real dates; and once with all of that hidden, just "day 1, day 2" with no names and no year.
>
> Think of it as a blind taste test. It's the same idea as a permutation test in statistics: if the labels carry no information, taking them away shouldn't change the result. And it didn't: accuracy was practically identical with and without the labels, 0.579 versus 0.583. Knowing it was Walmart in 2016 gave Claude no advantage.
>
> We also saved every response so the results can be re-checked, and checked that every answer came back in the right format. One honest limit remains: hiding the labels can't hide the sales pattern itself. If Claude had memorised the shape of these series, a blind test couldn't catch it, so contamination can't be fully ruled out.

Speaker note (background, not to read aloud): Our code asked Claude for 28 daily forecasts in a fixed JSON format, a list with one number per day. The risk is that a chatbot answers with prose, skips days or adds extras, which a program can't use. We required that format, checked each reply had exactly 28 days, and retried automatically if a reply failed the check. "224 requests" = 112 series (12 store × department totals + 100 items) × 2 versions (named and blind). All 224 came back usable. If asked: every reply is saved in our repo, so anyone can re-score Claude's forecasts without calling it again.

If asked about the permutation test: it is the same idea, not a formal permutation test. A real permutation test reshuffles the labels many times to get a p-value; we made one comparison, with labels versus without.

Point at: the dark caveat bar at the bottom.

Transition: "Speaker 4 will wrap up and show the dashboard."

Speaker 4 — answer, limitations and demo (3:30)

Slide 8 · Our answer: not yet on accuracy, yes on speed (1:45)

> So, can AI foundation models beat traditional methods? Our answer: not yet on accuracy, yes on speed. LightGBM, AutoETS and their simple average were the three most accurate. Chronos was the clear winner on speed: ten seconds, no training. And Claude matched the best traditional approach on store totals.
>
> Our results have limits. First, Claude was tested on a subset, and our blind test can't fully rule out that it saw this public data. Second, sales are not demand: when a shelf is empty, sales show zero even if customers wanted the product.
>
> And the most important limitation for a business: accuracy isn't the real goal. Ordering stock is a newsvendor problem with two different costs. If you order too little, you lose the margin on every sale you miss; that's the underage cost. If you order too much, unsold food is wasted at its cost; that's the overage cost. Because those costs depend on the margin, the best order isn't the forecast itself but a quantile of demand, and our accuracy metrics treat over- and under-forecasting the same. So what happened when we turned each model's forecasts into actual orders? For most products, it barely mattered which model we used: the profit differences were small enough to be luck. Only for high-margin products, where running out of stock is expensive, did the more accurate model, LightGBM, clearly make more money.
>
> So our recommendation: average a statistical model like AutoETS with LightGBM, use Chronos where speed or a lack of training data matters, and choose models on the metric that matches the business decision. Our next step is a rolling backtest over several windows.
>
> Let me show you the business problem in our dashboard.

Slide 9 · Demo: Shelf Signal dashboard (1:30 demo + 0:15 close)

Click the screenshot (or "Open live dashboard") in slide-show mode. If the internet fails, play the screen recording.

Time	Where to click	What to say
0:00–0:20	**Forecast explorer**: Level "Store x dept", any series	"This is Shelf Signal, which extends our teammate's original dashboard with all our models. Here's one store's department total: the black line is actual sales, and you can switch each model on and off. The ensemble and LightGBM follow the weekend peaks; Chronos runs low."
0:20–0:40	**AI lab**: scroll to the scatter, then "Read Claude's reasoning"	"Here Claude explains its own forecast in plain English, and this chart is the named-versus-blind test: every point sits on the diagonal, so real names gave it no advantage."
0:40–1:30	**Order planner**: drag the margin slider from 50% to 20%, then to 80%; point at the profit table	"Now the business problem. For this item, the model forecasts about eight units for the day. At a 50% margin, the best order is the median of demand. Drop the margin to 20%: a wasted unit now costs four times a lost sale, so the order falls. Raise it to 80%: a lost sale costs four times a wasted unit, so we order more. Same forecast, different decision. The table scores every model's orders against actual sales: the profit ranking isn't the same as the accuracy ranking, and at typical margins the differences are small."
Close	Stay on the dashboard	"That's our project: traditional methods still lead on accuracy, AI wins on speed, and the right order depends on your costs. Thank you, and we're happy to take questions."

Press Cmd+Tab to return to the slides for questions.

Before presenting

- [ ] Turn on GitHub Pages for the dashboard repo, then check that https://shihan-2002.github.io/shelf-signal-dashboard/ loads
- [ ] On the presenting laptop, open the dashboard once so it loads instantly; check the slide 9 link opens it
- [ ] Record a 90-second screen recording of the demo as a backup (QuickTime: File → New Screen Recording)
- [ ] Confirm in the M5 paper how many top teams used LightGBM before saying a number; until then say "dominated"
- [ ] Rehearse once with a timer; each speaker practises their transition line

Q&A prep

Likely question	Short answer
Why WRMSSE and not MAPE?	It is the official M5 metric, handles zero-sales days (MAPE divides by zero), and weights products by dollar sales.
Why is the ensemble better than LightGBM?	AutoETS and LightGBM make different kinds of errors, so averaging cancels part of them. Our teammate's Shelf Signal found the same.
Did Claude memorise the M5 data?	We can't rule it out. Hiding names and dates didn't change its accuracy (0.579 vs 0.583), but the sales pattern itself could still be recognised.
Was Chronos trained on M5?	No. The Chronos paper lists M5 among its held-out test sets, and it saw no retail data.
Why only 112 series for Claude?	Cost and time: about $0.06 per forecast, so all 5,748 series would cost about $347 and take hours.
Is your LightGBM as good as the M5 winners'?	No: one fixed setting and no ensemble of many models, so it is a conservative baseline.

Likely question	Short answer
Why do AutoARIMA and Chronos look bad on totals?	Summing nearly flat item forecasts loses the weekly pattern; fitted directly on the totals, both improve to about 0.64.
Is one 28-day window enough?	Not for firm rankings among close models. A rolling backtest over several windows is the next step.
What is the newsvendor problem?	Ordering under uncertain demand with two costs: the lost margin when you run out (underage) and the wasted cost of unsold stock (overage). The best order is the demand quantile equal to the margin.
Does better accuracy mean more profit?	Only sometimes: at typical margins the profit gaps were within chance; at an 80% margin the more accurate LightGBM earned significantly more.